

OpenH-RF Proposal:

The UltraSound ToolBox Channel Capture Collection

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1 Executive Summary

The UltraSound ToolBox (USTB)

The USTB is a free, open-source MATLAB toolbox for beamforming, processing, and visualization of ultrasonic signals, first released in 2017 [1]. It is developed at the University of Oslo in collaboration with NTNU, Université de Lyon, and Johns Hopkins University. USTB provides a unified processing framework based on the Ultrasound File Format (UFF, HDF5-based) and a General Beamformer architecture supporting all common transmit sequences (plane wave, diverging wave, focused, synthetic aperture, and retrospective beamforming). According to Google Scholar, the USTB has been **referenced in more than 150 publications**, spanning a wide range of topics such as adaptive beamforming, image quality assessment, deep learning for ultrasound, and clinical imaging research.

Proposed Contribution

We propose to contribute **51 USTB channel-capture datasets to OpenH-RF**. The collection is diverse in anatomy, hardware, and experimental and applied acquisition strategies:

- A In-vivo human (cardiac)** — parasternal long-axis and apical four-chamber imaging (3 acquisitions).
- B In-vivo human (vascular)** — carotid artery cross-section and longitudinal views (5 acquisitions).
- C Phantoms (Verasonics)** — tissue-mimicking CIRS targets for resolution, contrast, and dynamic range evaluation (18 acquisitions).
- D Phantoms (Alpinion)** — linear-array phantom benchmarks (4 acquisitions).
- E Simulations** — wave-propagation-based synthetic channel data (Field II): point scatterers, cysts, speckle, and blocked-array configurations (17 acquisitions).
- F Motion estimation** — elastography phantoms for shear-wave and ARFI push-tracking (4 acquisitions).

All data is openly hosted on Zenodo (<https://zenodo.org/records/20261898>). A catalog of all datasets, each with a reference Delay-And-Sum (DAS) beamformed B-mode image, is available at <https://unioslo.github.io/USTB/datasets.html>.

2 Task & Application Selection

Hardware

The experimental (non-simulated) channel data was recorded using two research ultrasound platforms, both providing full access to pre-beamformed RF channel data:

- **Verasonics Vantage 256** — majority of datasets (cardiac, vascular, phantoms, SWE/ARFI)
- **Alpinion E-Cube 12R** — phantom imaging with the L3-8 linear probe

Four transducer probes were used:

- **P4-2 phased array** (64 elements, Verasonics) — cardiac and focused imaging
- **L7-4 linear array** (128 elements, Verasonics) — vascular, phantom, and general imaging
- **L3-8 linear array** (128 elements, Alpinion) — phantom imaging (4 datasets)

Transmit Strategies

The collection covers five distinct transmit configurations: coherent plane-wave compounding (CPWC, 5–75 angles), diverging-wave imaging (DW, 25 virtual sources), focused single-line imaging (FI, 128 focused beams), synthetic transmit aperture (STA, 128 single-element transmits), and shear-wave push-tracking sequences.

Dataset Breakdown

ID	Application	No. of acq.	Tier	No. of frames	Probe / TX
<i>6.1 Generalized Reconstruction</i>					
A	Echocardiography	3	In-vivo (x40)	80	P4-2 / FI
B	Carotid imaging	5	In-vivo (x40)	8	L7-4 / FI, CPWC
C	Phantom (Verasonics)	18	Phantom (x1)	26	L7-4, P4-2 / CPWC, FI, STA
D	Phantom (Alpinion)	4	Phantom (x1)	4	L3-8 / FI, CPWC
E	Simulated	17	Simulation (x1)	22	Field II / CPWC, STA, FI
<i>6.4 Motion Estimation</i>					
F	SWE / ARFI	4	Phantom (x1)	800	L7-4 / push-track
Total		51		940	

Table 1: Dataset breakdown by application group (labels A–F match the executive summary). No. of acq. = number of acquisitions; No. of frames = total channel-capture frames across all acquisitions in the group.

Weighted contribution summary:

Tier	No. of datasets	Weight	Points
In-vivo human (research platform)	8	×40	320
Phantom / table-top	26	×1	26
Simulation (digital)	17	×1	17
Total	51		363

Weight denotes the OpenH-RF tier multiplier from the RFP (e.g. in-vivo human datasets on research platforms count 40× toward the contribution total). Points = No. of datasets × Weight.

3 Technical Approach

Data Format Conversion

Each UFF dataset is converted to OpenH-RF (zea HDF5) using a ready-made pipeline:

1. Read UFF with `pyuff-ustb` (channel data, probe geometry, transmit sequence).
2. Map to `zea` scan dictionary: `probe_geometry` [$N_{el} \times 3$], `polar_angles` [N_{tx}], `focus_distances`, `t0_delays` [$N_{tx} \times N_{el}$], `sampling_frequency`, `sound_speed`.

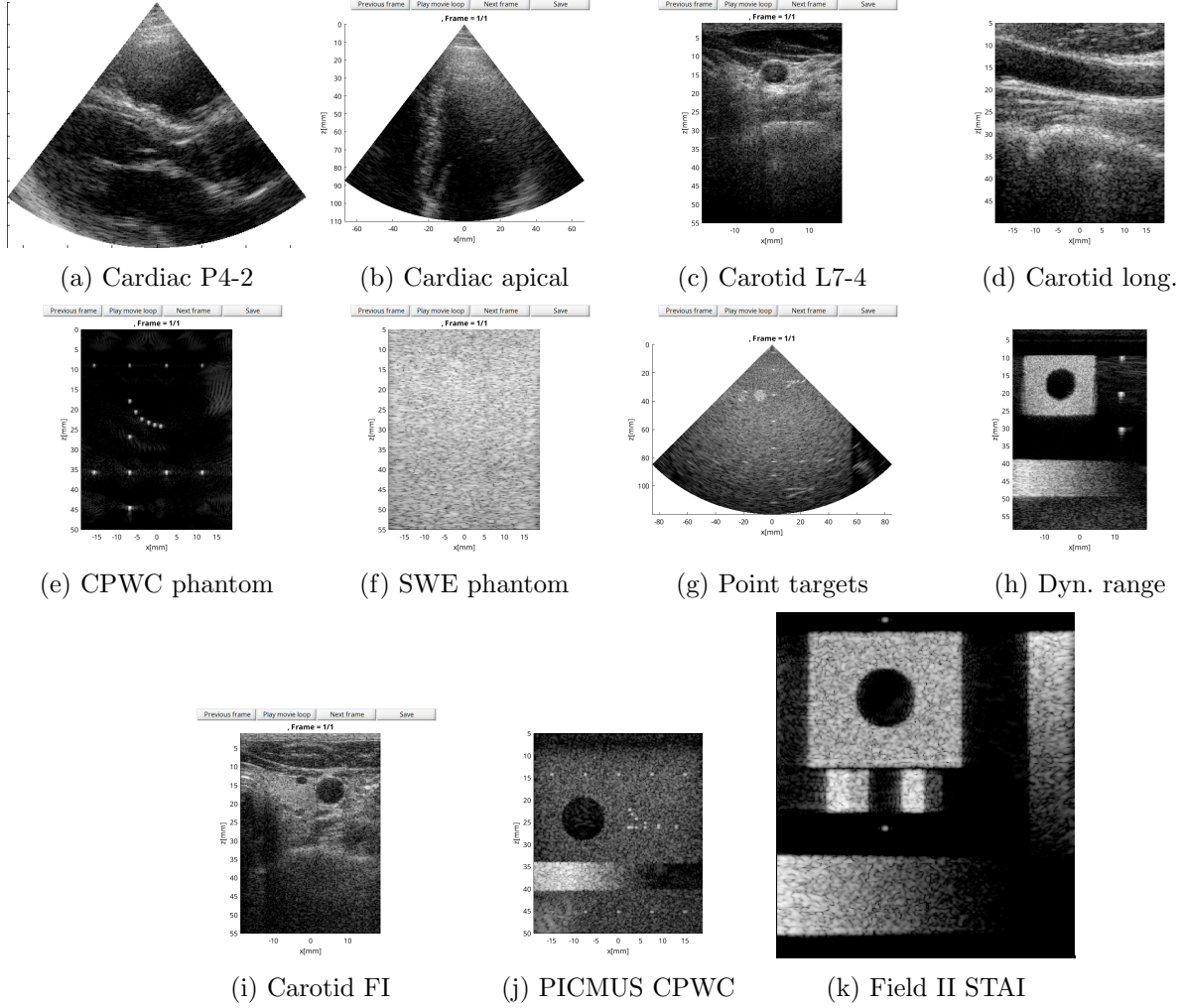


Figure 1: Representative Delay-And-Sum (DAS) reconstructions (60 dB dynamic range). Full catalog: <https://unioslo.github.io/USTB/datasets.html>

3. Generate Delay-And-Sum (DAS) B-mode image as reference reconstruction.

4. Write .hdf5 via `zea.File.create()`.

Both a Python script and a MATLAB equivalent are available at <https://github.com/unioslo/USTB/tree/master/sandbox/OpenHRF>.

Reference Reconstructions & Ground Truth

Delay-And-Sum (DAS) beamformed images ($f/1.75$, Hanning apodization, coherent compounding) serve as **reference reconstructions**. True ground truth is available where the scattering medium is fully defined:

- **Simulations:** Field II scatterer positions and medium parameters.
- **Phantoms:** manufacturer-specified geometry (CIRS Model 040GSE).
- **SWE/ARFI:** shear-wave velocity and displacement maps.

For in-vivo data, no ground truth exists; we supply raw channel data with **processing scripts for reconstruction** as permitted by the RFP (Section 6.7).

4 Data Governance, Ethics & Compliance

The in-vivo human datasets (cardiac and carotid) were recorded on healthy volunteers at the University of Oslo with **written informed consent** and approval from the **Regional Committee**

for **Medical and Health Research Ethics (REK)**. The UFF files contain only raw ultrasound channel data (backscattered RF signals) — no identifiable patient information is stored.

Phantom and simulation data have no human-subjects implications.

5 Data Rights & Licensing

All datasets are hosted on Zenodo under open-access terms. Code is released under the **MIT License**. We confirm intent to release the converted dataset under **CC BY 4.0** as required by OpenH-RF. No third-party IP encumbrances apply.

6 Team Qualifications

- **Ole Marius Hoel Rindal** — Senior Lecturer at UiO and co-founder of Sonair AS; lead developer of USTB since 2017; 15+ publications on adaptive beamforming, coherence-based imaging, and the Generalized Beamformer.
- **Yücel Karabiyik** — Chief Engineer and Facility Manager of the DSB ultrasound/sonar laboratory at UiO; PhD in Medical Technology (NTNU, 2017); responsible for programming and maintaining the Verasonics scanner and coordinating ultrasound data acquisitions.
- **Sven Peter Näsholm** — Professor and Head of the Digital Signal Processing and Image Analysis research group at UiO; acoustics, ultrasound signal processing, and medical imaging.
- **Andreas Austeng** — Professor and Head of the Machine Learning section at UiO; 20+ years in signal processing for medical ultrasound.

7 Project Plan

Task	Period	Deliverable
Finalize zea conversion pipeline	Completed	Scripts on GitHub
Generate reference reconstructions	Completed	51 DAS B-mode preview images
Submit proposal	Jun 10, 2026	PDF via Google Form
Convert all datasets to OpenH-RF HDF5	Jun 10–30, 2026	51 .hdf5 files
Deliver data	Jun 30, 2026	S3/GDrive upload

References

- [1] A. Rodriguez-Molares, O. M. H. Rindal, O. Bernard, A. Nair, M. A. L. Bell, H. Liebgott, A. Austeng, and L. Løvstakken, “The UltraSound ToolBox,” in *Proc. IEEE International Ultrasonics Symposium (IUS)*, Washington, DC, USA, 2017, pp. 1–4. doi: 10.1109/ULT-SYM.2017.8092389.